

MONALISA DOKANIA

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PROFESSIONAL SUMMARY

Full Stack software engineer focused on distributed systems, cloud infrastructure, and applied AI. I build fast, reliable, and thoughtfully designed production systems end-to-end. Currently a founding engineer, I also build & publish agentic AI systems.

TECHNICAL SKILLS

Languages	Python, Go, Java, TypeScript, JavaScript, HTML, CSS
Backend	FastAPI, Node.js, tRPC, Spring Boot, Flask, PostgreSQL, Redis, Kafka, Microservices
Frontend	React, Next.js, Vite, JavaScript / TypeScript, Tailwind, Vercel
AI / ML	Machine Learning, Agentic AI, Prompt Engineering, RAG, LangChain/LangGraph, Evals
Cloud & DevOps	AWS, EKS, Kubernetes, CI/CD, Docker, Terraform, Prometheus & Grafana
Others	Git, GitHub Actions, JIRA, Linux, AI tools (Cursor, Copilot, etc.), Testing frameworks

TECHNICAL EXPERIENCE

Lead Software Engineer BLUSVN Technologies	June 2024 – Present Scottsdale, AZ
• Founding engineer on a distributed water-delivery platform; building from 0 to 1, and owning the full product lifecycle from architecture through production across backend, infrastructure, and ML. • Architected and shipped 100+ TypeScript/tRPC backend APIs powering driver and customer React apps, including identity and auth workflows; Improved data-fetching efficiency and reduced end-to-end latency by 40%. • Engineered high-performance Golang microservices with concurrent worker pools and batched notification delivery, designed for high-throughput driver allocation and order processing at scale. • Built real-time ML inference pipelines (YOLO) for face and ID validation, and AI-assisted agent workflows (multi-agent system) for document summarization and log analysis/reporting that cut manual triage time on production incidents. • Designed and operated production AWS/EKS infrastructure with Terraform; migrated from Cluster Autoscaler to Karpenter for faster node provisioning and reduced AWS spend by 12%. • Hardened security posture with AWS/EKS best practices - cut high-severity CVEs by 20%, cluster compliance from 65% to 76%. • Built Kafka-based event-driven location workflows to support responsive UI experiences in order & delivery status tracking. • Led load and chaos testing with Locust and fault-injection drills, surfacing bottlenecks and validating SLOs before production rollout - validated the platform at 50K orders/min, and monitored via Prometheus + Grafana observability stack. • Partnered with product managers & stakeholders on technical architecture; led engineering delivery; managed CI/CD & sprints.	
Graduate Teaching Assistant (GTA/GSA) Arizona State University (SCAI)	May 2023 - May 2024 Tempe, AZ
• Supported Software Verification, Validation Testing (900+ students across sections). Led lab sessions and office hours. • Guided students in unit testing, debugging, code reviews, and OOP principles. Developed course materials and grading rubrics.	

PROJECTS

Agent-Based AI System for Document & Log Analysis 🐙 <i>Python, LangChain, FastAPI, Redis RQ, Vector DB</i>
• Built a multi-agent LLM analysis pipeline (Planner → Retriever → Analyst → Critic → Reporter) for log analysis & reporting. Implemented a RAG workflow for semantic document retrieval for document analysis using FastAPI and Qdrant vector store.
Research Assistant — Hybrid RAG + RAGAS Evaluation 🐙 <i>Python, React/TypeScript, RAG, LangChain, ChromaDB</i>
• Document research assistant built in four parts - naive baseline → advanced retrieval (hybrid search, cross-encoder re-ranking, citation enforcement) → RAGAS eval against 60 ground-truth Q&A pairs → systematic prompt iteration with measured improvements.

EDUCATION

Master of Science, Computer Science Arizona State University Tempe, AZ	GPA: 4.0/4.0
Bachelor of Technology, Information Technology Motilal Nehru National Institute of Technology Allahabad, UP, India	CGPA: 7.59/10.0